PHD125 Pharmaceutical Organic Chemistry · Carbohydrates

Before class: sugars, rings and mirror images

Dr. NAM · Faculty of Pharmacy, UiTM Bertam, Penang

About 15 minutes. Read the three short cards, then try the readiness check. Wrong answers are fine: they show what to listen for in the lecture.

Read the cards first, then answer the questions without notes. The answer key is at the back.

1 · Read first (5 min)

One building block

Glucose, $C_6H_{12}O_6$, is an aldehyde with an –OH on almost every carbon. Plants make it from CO_2 and water using sunlight, then join glucose units into **starch** (food store) and **cellulose** (fibre). Table sugar is glucose joined to fructose.

Left and right hands

In Chapter 2 you met the **chiral carbon**: a carbon with four different groups. Such a molecule has a mirror image that cannot be placed on top of it, like your hands. Glucose has four chiral carbons. Enzymes and drug receptors can tell the mirror images apart.

Sugars curl up

In Chapter 3 you saw that an aldehyde reacts with an alcohol to form a **hemiacetal**. Glucose contains both groups in one molecule, so in water it bends round and forms a **ring**. The ring can close in two ways, called α and β, and that small difference decides whether we can digest a carbohydrate.

2 · Readiness check

Ten quick questions: recap and a preview.

Q1 Recap

A chiral carbon is bonded to:

A. four different groups
B. two identical groups
C. a double bond
D. four hydrogen atoms

Q2 Recap

An aldehyde reacts with an alcohol to form a:

A. hemiacetal
B. ketone
C. carboxylic acid
D. ether

Q3 Recap

Benedict's reagent turns brick-red with:

A. aldehydes
B. ketones such as acetone
C. alcohols
D. ethers

Q4 CH5-LO1

The molecular formula of glucose is:

A. $C_6H_{12}O_6$
B. $C_{12}H_{22}O_{11}$
C. C_6H_6
D. CH_2O

Q5 CH5-LO5

Match each carbohydrate to its type.

Options: monosaccharide · disaccharide · polysaccharide

glucose _____

sucrose _____

starch _____

Q6 CH5-LO1

Two mirror-image forms of a drug always have the same effect in the body.

True / False

Q7 CH5-LO2

Glucose contains an aldehyde group. It is therefore called an:

A. aldose

B. ketose

C. alditol

D. acetal

Q8 CH5-LO5

Which sugar is in milk?

A. lactose

B. sucrose

C. maltose

D. fructose

Q9 CH5-LO5

Which do humans NOT digest?

A. cellulose
B. starch
C. sucrose
D. glycogen

Q10 CH5-LO5

Many tablets contain lactose or starch as fillers. Suggest one reason a pharmacist might check the filler before dispensing.

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About 15 minutes. Read the three short cards, then try the readiness check. Wrong answers are fine: they show what to listen for in the lecture.

Answer key

2 · Readiness check

Q1 Recap

A chiral carbon is bonded to:

A. four different groups

B. two identical groups

C. a double bond

D. four hydrogen atoms

Chapter 2.

Q2 Recap

An aldehyde reacts with an alcohol to form a:

A. hemiacetal

B. ketone

C. carboxylic acid

D. ether

Chapter 3: this is how sugar rings form.

Q3 Recap

Benedict's reagent turns brick-red with:

A. aldehydes

B. ketones such as acetone

C. alcohols

D. ethers

Cu^{2+} is reduced to Cu_2O (Chapter 3).

Q4 CH5-LO1

The molecular formula of glucose is:

A. $C_6H_{12}O_6$

B. $C_{12}H_{22}O_{11}$

C. C_6H_6

D. CH_2O

$C_{12}H_{22}O_{11}$ is a disaccharide such as sucrose.

Q5 CH5-LO5

Match each carbohydrate to its type.

glucose → monosaccharide; sucrose → disaccharide; starch → polysaccharide

One, two or many sugar units.

Q6 CH5-LO1

Two mirror-image forms of a drug always have the same effect in the body.

True / False → **False**

Thalidomide is the famous example: receptors and enzymes are chiral.

Q7 CH5-LO2

Glucose contains an aldehyde group. It is therefore called an:

A. aldose

B. ketose

C. alditol

D. acetal

Aldehyde → aldose; ketone → ketose.

Q8 CH5-LO5

Which sugar is in milk?

A. lactose

B. sucrose

C. maltose

D. fructose

Lactose, milk sugar.

Q9 CH5-LO5

Which do humans NOT digest?

A. cellulose

B. starch

C. sucrose

D. glycogen

We lack an enzyme for its β links.

Q10 CH5-LO5

Many tablets contain lactose or starch as fillers. Suggest one reason a pharmacist might check the filler before dispensing.

A patient may be lactose intolerant, have galactosemia, follow a diet, or have diabetes and want to avoid sugars; some patients also avoid certain starches (for example gluten-containing wheat starch). Fillers are part of what the patient swallows.

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